

Smart Inventory Manager

Wrong inventory **kills margins but**, Smart inventory **multiplies them**.
we predict exactly what to order, when, and how much

AI Capstone Project

The Problem & Market Opportunity

Inventory planning is still driven by guesswork — not data.



Every Reorder is a Guess

Managers order based on intuition with no system to predict how much is needed or when



Dead Stock = Dead Capital

42% of SMB retailers have excess inventory that ties up cash and generates zero return



Stockout Loses the Customer

91% of customers won't return after a stockout — you don't just **lose the sale**, you **lose the customer**



Zero Early Warning

No system detects demand shifts early — by the time you react, shelves are empty or capital is already frozen

\$491M → \$4.07B

Egypt AI market — 42% CAGR to 2031

Source: Statista Market Forecast, 2025

65%

Reduction in stockout sales losses with AI Forecasting

Source: McKinsey & Company — Supply Chain AI Impact

20 – 30%

Reduction in overall inventory levels via ML forecasting

Source: McKinsey & Company, Nov 2024

\$1B → \$4B

AI in Retail — **MEA** region, **26.6%** CAGR to 2030

Source: Grand View Research, 2024

\$4.5B → \$14.6B

AI spending in META region — **34%** CAGR to 2028

Source: IDC Worldwide AI Spending Guide, 2024

20 – 50%

Reduction in demand forecasting errors + 5–10% lower storage costs with AI

Source: McKinsey & Company, Nov 2024

The AI Solution

From guesswork to guaranteed decisions



Every item on your shelf is either **making you money** or **costing you money**. Now you'll always know which.
know which.



90-Day Forecast

Forecasts with 3 confidence levels

(**best case**, **expected**, **worst case**)

so your reorder decision has a risk estimate, not just a
not just a number



Decision Engine

Recommends quantities per product per branch,
branch, flags the lower-risk supplier, and alerts you
alerts you 7–14 days before a stockout



Multi-Branch Intelligence

When one branch has excess and another runs
runs short, the system recommends an internal
internal transfer — no new order needed



Automatic Alerts

Connect your data once, set your schedule — **daily**,
weekly, or **monthly** — and get Arabic email alerts
automatically



Seasons Don't Surprise It

Built-in calendar for Ramadan, Eid, Back to School,
and Black Friday — extra stock added per product
automatically



Report & Order in few Min

Arabic executive report and a ready-to-send Purchase
Order PDF — generated automatically, no analyst
needed.

How It Works — The AI Pipeline



No setup week — works immediately for new customers with zero historical training



Drift Monitoring & Alerts with Low / Median / High confidence bands per (store × item)



Dynamic Safety Buffers Smart reorder quantities adapt to each product's volatility

Service Level & Smart Decisions

From operational metrics to boardroom language



Service Level KPI

Current Service Level

82%



Predicted Service Level

96%

Every buffer tier maps to a concrete service-level target: 10% buffer $\approx$ 87% service, 20% $\approx$ 93%, 35% $\approx$ 97.5%. Management speaks in fill rates — now your AI does too.



What-If Simulation

*What if demand increases by **30%**?*

3 products at risk

+450 units

+EGP 45K

Simulate any demand scenario instantly. See which products are at risk, how much additional stock you need, and what it costs — before committing capital.



ABC Analysis

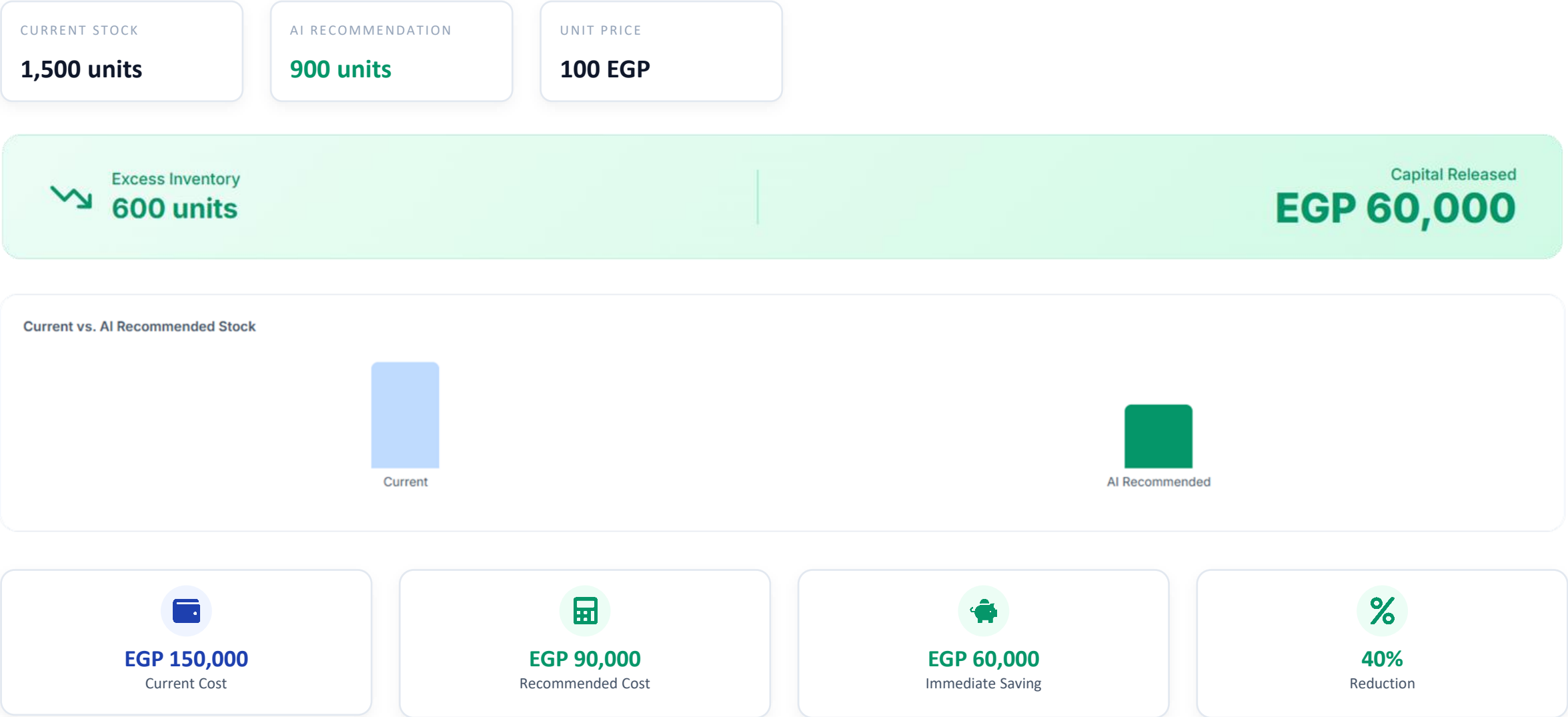


A = Highest value B = Medium value C = Low value

Pareto (Products automatically classified by value) contribution. Category A gets priority alerts and tighter monitoring — focus where it matters most.

Inventory Cost Impact — Real Numbers

See exactly how much capital our AI can release



The Transformation

What changes the day you turn on Smart Inventory Manager

✗ BEFORE

Inventory as a Cost Center

- ✗ Manual spreadsheets updated weekly — decisions delayed by days
- ✗ Safety stock calculated with one generic formula for all products
- ✗ No warning until the stockout actually happens
- ✗ Purchasing based on gut feel — no data-driven supplier comparison
- ✗ Excess capital locked in overstock with no visibility into how much
- ✗ Reports take 3+ hours to compile — if they get done at all

✓ AFTER

Inventory as a Profit Decision

- ✓ Real-time AI recommendations updated automatically — zero manual work
- ✓ Dynamic safety buffers per product based on actual demand volatility
- ✓ Early alerts 7-14 days before stockout — time to act
- ✓ Supplier risk scores recommend the best option every time
- ✓ Exact capital-release calculation per SKU: EGP 60,000 freed in example
- ✓ Executive report generated in under 2 minutes — consulting-grade quality



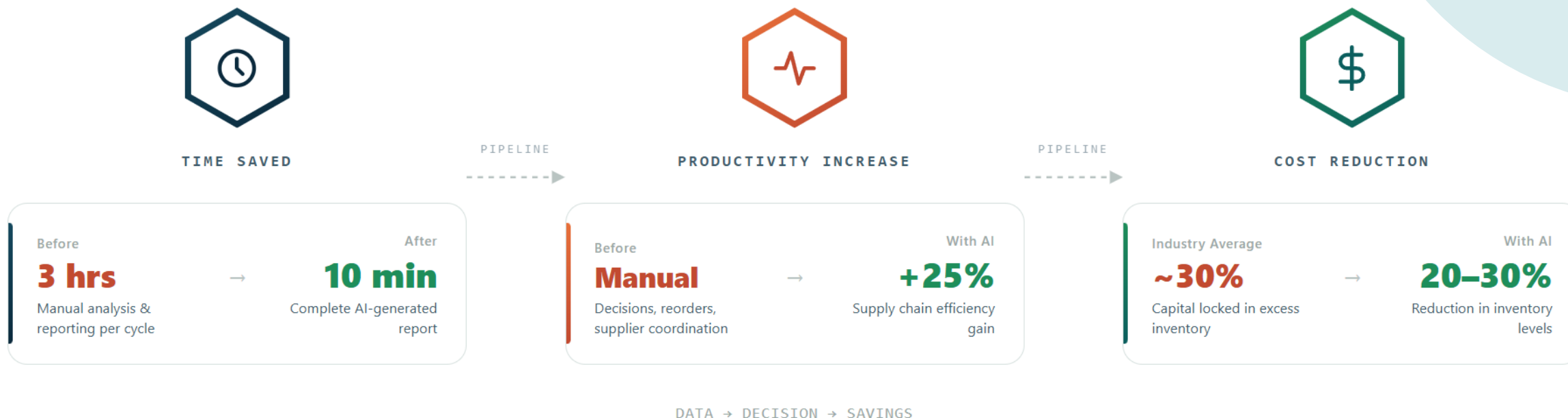
Scan to Watch Demo

Demo Mode Available

Pre-loaded sample data guarantees every live presentation runs flawlessly.

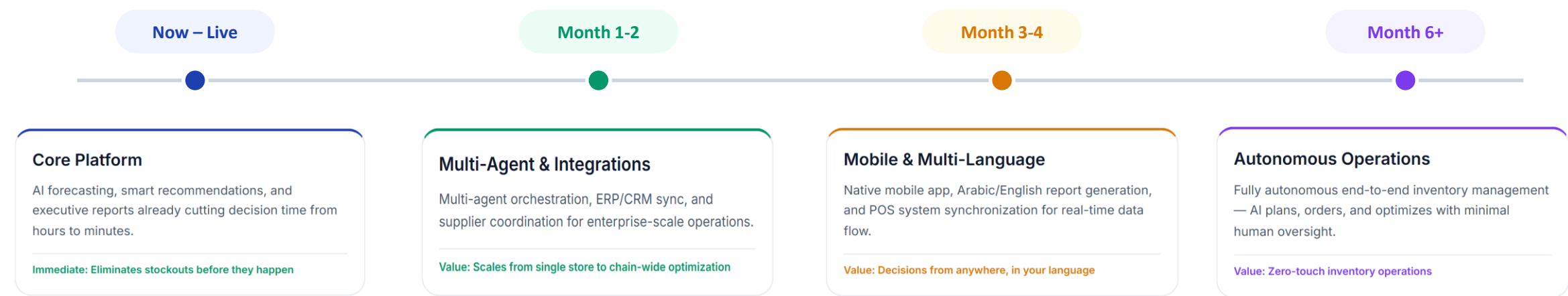
Projected Business Impact

Measurable outcomes, not just technical improvements



Our Roadmap

Every phase delivers measurable business value



PROJECT SUMMARY

An AI system that turns raw inventory data into ready-to-act operational decisions — forecasting demand, detecting drift, and writing its own reports without human intervention.

MAIN BUSINESS VALUE

It turns 3 hours of manual analysis into a ready decision in 10 minutes — available to any number of users, anytime, without downtime.

FINAL MESSAGE

Because the best systems aren't demoed — they're built to run. And this one runs.

Thank You

Questions & Discussion

Smart Inventory Manager



Mahoud.Alkodousy@gmail.com



smartinventorymanager.com



github.com/MahmoudAlkodousy/smart-inventory-manager



Scan for Demo

Turning inventory from a **cost center** into a **profit decision**.